



ages are unacceptable! Currently, power is derived from the biggest power source in our solar system—our sun. Even the two nomad rovers that are wandering about exploring the Martian surface are solar powered...

The solar panels used on the rovers are not your ordinary run-of-the-mill solar cells; they're tri-layered cells stacked one on top of the other to absorb every last ray of solar power that they can. Currently the two rovers carry two 8 amp-hour batteries that are charged using the solar panels. Each rover was able to produce between 400 and 900 watt-hours of power per Martian day—that's enough to power a 100 watt light bulb between 4 and 9 hours! Of course, when designing all electrical systems and electronic components, NASA's engineers are notably stingy with power consumption values—which results in such a reduced load that the power generated by the solar cells is quite sufficient (unless an astronaut decides to plug in his boom-box for some entertainment!).

The drawback of solar power is that it requires rovers or astronauts to remain in sunlight-rich areas—of Mars or otherwise. This is generally the equatorial region, and trips to the bottom of Martian chasms are a strict no-no. Alternative power generation techniques are being researched, and it's quite probable that the first human exploration of Mars will be nuclear-powered.

Communications

A lot of mankind's space exploration attempts have ended in disaster. A lot of us know about the Challenger disaster in 1986, where seven astronauts were lost. More recently, space shuttle Columbia, with Indian-born Kalpana Chawla onboard, broke up on re-entry in February 2003. Such are the risks associated with human space flight, and these are examples of the worst type of disaster. However, a seemingly smaller and more nagging problem lies in the communication systems. This is more pronounced in unmanned flights, where computers cannot just fix themselves, and often the end result is loss of the spacecraft. Just last year, in November 2006, NASA lost contact with their Mars Global Surveyor, a craft that had been orbiting Mars since 1997. It's still there, orbiting Mars, but NASA just cannot contact it or get it to respond.

This is perhaps the biggest risk of unmanned flights, and while it does not involve loss of human life, billions of dollars in research and development can go down the drain if one antenna fails to point in the right direction. Communication is the key, and it can spell the difference between a miracle man-made robotic explorer and good old space junk.

In order to stay in constant touch with all the spacecrafts out there in space, NASA has what they term the "Deep Space Network" (DSN). This consists of three high-tech communication facilities—one in California's Mojave desert, another near Madrid in Spain and the last one near Canberra in Australia. Working in tandem, these antennas, the most powerful and sensitive telecommunications equipment known to man, can communicate with all working space probes and crafts that have been sent out, no matter where they are.



Super Solar Sail

The rovers took seven months to reach Mars from Earth on conventional propulsion techniques. Drawing inspiration from science fiction, there are many alternative propulsion techniques that authors have written about. Some will stay fiction forever, one is already a reality. Introducing the solar sail, which science fiction works tell us is a huge sail attached to a spaceship that uses the light (photons) emitted from the sun as a "solar wind," propelling the craft attached to it away from the sun. Traditionally, sci-fi writers have spoken of solar sails as engineless, unmanned spacecraft. This would be a good solution for sending a probe out to another solar system, for example, because, honestly, we don't care how long it takes.

You've heard of the Wright brothers... get ready for the Benford twins from Alabama. In 2005, the brothers developed a sail that they claimed could have gotten the two rovers to Mars in under a month! The principle is to not depend on the sun as the energy source, but to generate our own "wind". They created a sail made of a carbon mesh that they hit with directed microwaves. The result was that carbon monoxide was being given off and the sail experienced a considerable "push". They then worked on the theory that a sail could be painted with a film of specially created paint that, when heated, gave off gas and propelled the sail (and the craft attached to it) away.

According to their calculations, if a spacecraft in Earth orbit were to unfurl a square sail 100 metres across, and was bombarded for an hour by a 60 megawatt microwave, the spacecraft would accelerate to a speed of 60 kilometres per second—faster than any spaceship has ever gone. This would enable a ship from Earth to reach Mars in under a month.

Now, 60 megawatts is a *lot* of power, and though a lot of people are working on building sails, none that can be folded or stored are possible yet. Perhaps the future will see a combination of the solar sail and nuclear-powered propulsion to get Man to Mars.

As for the spacecraft out there in space, communication is taken care of by two main antennas—a low gain, omni-directional antenna and a mid- / high-gain directional one. The low gain antenna is used when the spacecraft is close enough to Earth to not require focused communicating rays, but once the distance starts increasing to the order of a few million km, the mid-/high-gain antenna is needed to stay in touch with Earth. The drawback of directional antennas is that they are motorised, so there's a highly increased risk of failure. In fact, just before they finally lost contact with the Mars Global Surveyor (MGS), it communicated to NASA that it had a mechanical problem with a solar panel, and rotated itself to make sure it got enough sunlight. However, this rotation pointed the high-gain antenna away from Earth and also exposed the battery to direct sunlight, heating it up beyond